

QUIC Performance Evaluation On a Wifi Network With NS-3 Simulation

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Abstract—QUIC (Quic UDP Internet Connection) is a communication protocol that handles the tasks and responsibilities of the transport layer. QUIC has been standardized by the Internet Engineering Task Force (IETF) since 2021 and aims to solve the inherent problems of the TCP (Transmission Control Protocol) protocol so as to improve data transmission performance. In this final project, a series of experiments will be carried out on the QUIC and TCP protocols in the scenario of using the Internet via WiFi channels. The purpose of this experiment is to review and compare the characteristics of the QUIC protocol against TCP. Two experimental scenarios were tested, namely, testing both protocols independently and simultaneously. To carry out this series of experiments, NS-3 network simulation software was used. The results of the simulation show that QUIC shows an increase in throughput values compared to TCP, but QUIC shows jitter, packet loss, and delay with higher deviations.

Index Terms—QUIC, Simulation, HTTP/3, Wifi, NS-3

I. INTRODUCTION

Today, most of the Web runs using the hypertext transfer protocol (HTTP) at the application layer and the Transmission Control Protocol (TCP) protocol at the transport layer. TCP was first published in 1981 through RFC 793 and HTTP in 1996 through RFC 1945 [1] [2]. Since then, these two protocols have become the main protocols that support more than 90 percent of global internet traffic [3].

HTTP has undergone 3 version updates that gave birth to HTTP/1.1, HTTP/2, and most recently, HTTP/3. One of the most striking differences between HTTP/3 compared to previous versions is the use of a new technology called QUIC, which is encapsulated in the user datagram protocol (UDP) instead of using TCP at the transport layer. Other improvements include more efficient header compression and transport layer security version 1.3 (TLS 1.3) integrated with QUIC [4]. An overview of this technology and protocol stack can be seen in figure 1.

QUIC brings new features that can overcome the shortcomings of TCP while maintaining TCP functionality, such as error retransmission, acknowledgment, and congestion control. The encapsulation of QUIC in UDP aims to make this protocol run and be understood by middleboxes and infrastructures connected to the internet before the QUIC protocol [6]. If QUIC is directly run without UDP encapsulation, all middleboxes on the internet must be configured to support this protocol. This is certainly very troublesome and takes a long time.

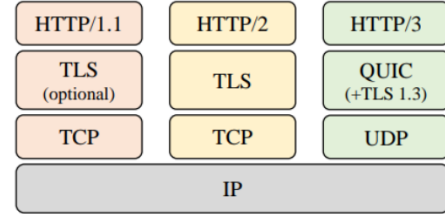


Fig. 1. HTTP protocol layer stack [5]

QUIC solves TCP's inherent shortcomings of head-of-line blocking (HoL) and long connection establishment duration. A head-of-line problem is when there is a disturbance at the end of data transmission that prevents the next data from being sent until the disturbance is resolved. This problem is a common phenomenon in computer network systems. This phenomenon originally also occurred in the HTTP/1.1 protocol at the application layer. Then, HTTP/2 overcomes this problem by using a multiplexing mechanism so that data can be sent more independently so that interruptions in data transmission do not inhibit the transmission of other data segments. Unfortunately, this solution only solves the HoL problem at the application layer (HTTP), while the layer below it (TCP) also has HoL problems. TCP is a protocol that maintains connections and has a retransmission guarantee for lost or problematic network packets. In this case, another packet process must wait for the retransmission process before it can be sent again. This can lead to HoL in the TCP protocol. Just as HTTP/2 addresses the HoL problem at the application layer, HTTP/3 uses the QUIC protocol to address the HoL problem at the transport layer.

II. RELATED WORK

Research publications by Apostolos I. Kyratzis and Panayotis discuss the comparison and performance improvement between QUIC and TCP on LTE wireless connections [6]. Simulations are run according to 4 experimental scenarios, namely steady-state throughput, flow fairness, file downloads, and the effect of QUIC streams on throughput. The steady-state throughput scenario tests the performance of each protocol in the case of constant data transmission and downstream transmission direction. Furthermore, the flow fairness scenario aims to

examine the fairness of bandwidth sharing of one network traffic to other network traffic passing through the same media. To do this in the LTE network, the same bulk send transmission pattern as the first scenario is used with variations in the combination of the number of user equipment and the type of protocol it runs. Furthermore, the file downloads scenario aims to examine the download duration of QUIC and TCP with a variation of 64 kB - 2 MB file size. Finally, the fourth scenario aims to test the new feature presented by QUIC, which is stream multiplexing. This test aims to increase the throughput against the variation of the number of streams used.

The research publication by Xiang Qin et al discusses the performance comparison of TCP and QUIC over the Internet in a real-world environment [7]. Experiments in this research directly use hardware consisting of clients (mobile devices and personal computers), controlled servers, and actual servers. The application variations used in this test are web browser and online video. This experiment consists of two stages, namely collecting network traffic data on the testbed (controlled server) and the actual network (actual server). Then, the next stage is computing the collected data to analyze the performance of QUIC and TCP. From the collected data, four aspects of the research were analyzed, namely the comparison between QUIC and TCP in web browsing and online video, fairness between QUIC and TCP, and the effect of the 0-RTT feature on QUIC.

Based on the experiments and findings, this research concludes that QUIC generally outperforms TCP in various experiments and this is due to the characteristics possessed by QUIC. The most influential characteristics are the elimination of head-of-line blocking and the reduction of connection handshaking duration, such as the 1-RTT and 0-RTT features. QUIC is also more reliable in poor network conditions when paired with delay-based congestion control algorithms such as BBR.

III. METHODOLOGY / SIMULATION SCENARIO

The method to be used in this final project is simulation using network simulation software. The simulation method was chosen due to the limited time and resources of the research. The simulation software that will be used is NS-3 Network Simulator version 3.39 because it is a network simulation that has all the modules needed to simulate this final project research. The NS-3 simulator is a code-based simulator using the C++ language as its framework and is based on object-oriented programming (OOP). So, the combination of complete modules and configuration that can be done in a structured manner through OOP code can allow for detailed scenario configuration with a high level of control. In addition, this simulator has been widely used in other studies on the same topic.

A. Initial Study Experiment

The initial study experiment aims to determine the test baseline and ensure that the simulation configuration can be run as expected. This simulation scenario aims to determine the

effect of distance between AP (Access Point) and STA (Station) on the maximum throughput that can be generated. The simulation is configured according to the topology illustrated in Figure 2. There are 4 node devices in this simulation, namely, wifi station (client), wifi access point, gateway, and server. The variable values and other parameters are set according to table I.

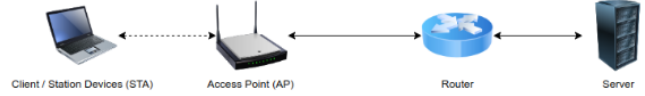


Fig. 2. Baseline topology

TABLE I
INITIAL EXPERIMENT CONFIGURATION

Variable	Configuration
Distance from STA to AP	1-60 meter
Simulation duration	60 second
Signal propagation loss model	ns3::FriisPropagationLossModel
Saluran nirkabel	IEEE 802.11n (wifi 4; 2.4 GHz)
Wifi rate algorithm	ns3::MinstrelHT
Wire connection	point-to-point
Wire connection bandwidth	1Gbps
Wire connection delay	2ms
Wire connection error rate	0%
Data transmission rate	100Mbps
Maximum packet size	1420 byte
Data transmission direction	upstream dan downstream
Transport layer protocol	UDP

Instead of using the QUIC or TCP transport protocols, this simulation uses the UDP transport protocol. This aims to minimize the effect of protocol-level bottlenecks such as flow control and congestion control mechanisms. The data generated in this simulation will be used to compare TCP and QUIC protocols as the best scenario.

B. TCP And QUIC On Independent Connection

Testing TCP and QUIC independently aims to test the performance of both protocols when running independently. The test scenario, topology and variables used follow the baseline test conducted in the initial study experiment stage. The variables and parameters used are set according to the configuration and topology used in the previous section. However, instead of using UDP, the protocols used were TCP and QUIC with each running in a separate simulation.

TABLE II
INDEPENDENT SCENARIO CONFIGURATION

Variable	Configuration
Congestion control algorithm	ns::TcpNewReno
Transmission pattern (application type)	on-off and bulk send
On-off duration	1 second
Maximum byte sent	Unlimited
Transport protocol	TCP and QUIC

The additional variables presented in this simulation include congestion control algorithms and on-off and bulk-send transmission patterns. These variables can be seen in table II. The New Reno algorithm was chosen due to its popularity and its wide use on various network devices. Meanwhile, on-off and bulk-send transmission patterns are features provided by NS-3 and are widely used in various studies to simulate Internet usage. The on-off application has a data transmission pattern that is on (sending data) and off at a certain duration that can be set. This feature aims to simulate internet usage with a browsing model. Bulk send is a transmission pattern that resembles the file upload or file download process. The bulk send application will try to send chunks of data as quickly as possible. However, in this simulation, the size of the sent data chunks is not limited. This is to know the maximum capability and throughput that can be achieved.

C. TCP And QUIC On Shared Connection

This part of the experiment aims to determine the throughput value of QUIC and TCP when running on the same network and in the same or adjacent time range. Based on previous experiments, it is known that variations in transmission direction and application type do not make a significant difference to the throughput value. In addition, it is also known that the optimum distance between AP to STA that shows a significant difference in throughput between protocols is 1-25 meters. Therefore, the experiments in this section focus on the comparison of throughput between protocols on the same network and constant distance. This experiment also aims to observe the phenomenon of "bandwidth seizure" between protocols in a wireless network. From previous research, it is known that in wired networks, QUIC tends to get more bandwidth than TCP [8]. Therefore, the topology used is a development of the previous subchapter experiment, namely the dumbbell topology with a wifi access point at one end. The difference from the previous topology is that there are 2 pairs of STA devices and servers running using different protocols but still through the same network. This is depicted in figure 3.

This experiment was conducted in scenarios, and each scenario consisted of three sub-scenarios

- 1) TCP and UDP:
 - a) TCP and UDP run at the same time
 - b) TCP runs first, followed by UDP
 - c) UDP runs first, followed by TCP
- 2) QUIC and UDP:
 - a) QUIC and UDP run at the same time
 - b) QUIC runs first, followed by UDP
 - c) UDP runs first, followed by QUIC
- 3) TCP and QUIC:
 - a) TCP and QUIC run at the same time
 - b) TCP runs first, followed by QUIC
 - c) QUIC runs first, followed by TCP

It should be noted that the function of UDP in this series of experiments is as a comparison protocol. This scenario was run using the same variable value configuration as the previous

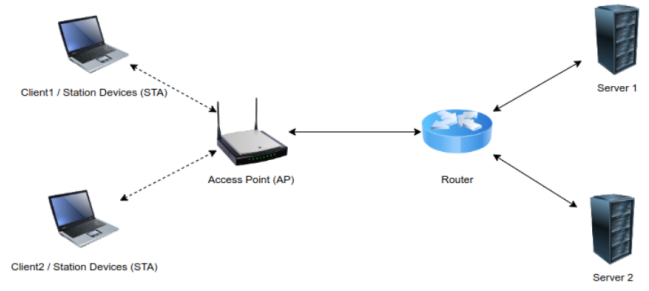


Fig. 3. Dumbbell topology

experiment. However, based on the temporary conclusions obtained, the test was limited to a distance of 5 meters and 15 meters, which is in the range of optimum AP to STA distance. For the test to better capture the phenomenon of bandwidth contention, the test time was extended from 1 at one distance to 20 seconds for each distance. For a series of scenarios where two protocols are not running simultaneously, a 10-second start pause is given between protocols. Meanwhile, the application and transmission direction used are on-off applications with an upstream direction.

IV. ANALYSIS AND RESULT

A. Initial Study Experiment

Figure 4 shows the average throughput data and signal strength received by wifi devices against the distance in the upstream and downstream scenarios. The average throughput speeds obtained in the upstream and downstream scenarios are 40.737 Mbps and 40.761 Mbps. It was found that the decrease in throughput in the upstream and downstream scenarios has the same trend and occurs at the same distance and time. So, it can be concluded that the upstream and downstream variations in this simulation do not produce significant differences in throughput.

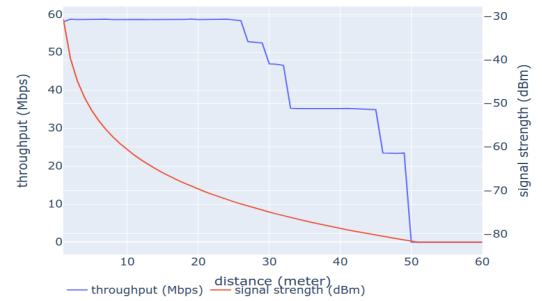


Fig. 4. Signal vs Throughput

Figure 5 also shows that the simulation program successfully simulates signal propagation according to the Friis equation so that the signal strength is obtained, which shows an exponential downward trend. The trend of decreasing throughput that resembles a stepped staircase shows the effect of the wifi rate algorithm for adjusting the optimal bitrate on decreasing the received signal strength. It can be seen that there are three

levels of throughput that decrease at a distance of 1-25 meters, 33-45 meters, and 46-49 meters before finally falling to zero. This stepped ladder pattern resembles the research conducted by I. Sammour et al [9].

B. TCP And QUIC On Independent Connection

Experimental results show that the protocol has a trend of decreasing the throughput of the stepped staircase, which also occurred in the initial study experiments. The decrease in throughput also occurs at the same distance as the initial study experiment. The average throughput results in this experiment are listed in Table III. The data in table III shows that QUIC has a higher throughput value in all application scenarios and transmission directions. From this table, it is also known that the throughput value is more influenced by the transmission direction than by the application type.

TABLE III
INDEPENDENT CONNECTION RESULT

Application Type	Direction	Avg. Throughput	
		TCP (Mbps)	QUIC (Mbps)
On-off	Upstream	31.71	34.86
	Downstream	33.21	34.49
Bulk send	Upstream	31.50	35.17
	Downstream	33.21	34.88

TABLE IV
INDEPENDENT CONNECTION RESULT (AVERAGE THROUGHPUT)

Distance	TCP	QUIC	UDP	Standard Deviation
1-25 meter	45.79	50.85	58.62	5.27
33-45 meter	27.78	30.80	35.12	3.01
46-49 meter	18.52	20.75	23.43	2.00

Table IV shows the average throughput value of each protocol over a range of wifi rate distances according to the experimental results of the initial study. For TCP and QUIC, the average throughput is obtained from the aggregation of both application types and transmission directions. From the results obtained, it is known that in all distance ranges, QUIC has a throughput value that is between TCP and UDP. This is in line with the expectation, considering that QUIC basically runs on top of UDP protocol but with all the enhanced features owned by TCP. It can also be seen that there is a downward trend in the standard deviation of the three, indicating that the significance of the throughput value between the three decreases as the distance increases.

The results of this experiment also show that the difference in protocol type has more influence on the trend and jitter value than the difference in application type. Qualitatively, it can be seen that the TCP protocol has better jitter than QUIC. This can be seen from the more stable trend of data deviation in TCP data, while QUIC has a wider and more contrasting data deviation. The average jitter obtained from the combined scenarios is 1843 milliseconds for TCP and 1852 milliseconds for QUIC. The standard deviation of the jitter value for TCP is 33 milliseconds, and it is 99 milliseconds for QUIC. This

shows that although TCP has a smaller average bandwidth than QUIC, the connection with TCP is more stable than QUIC.

In addition, the results show the delay values of TCP and QUIC with varying transmission directions and application types. It was found that QUIC has a more noticeable difference between upstream and downstream transmission compared to TCP, which is relatively the same. TCP has an average delay duration of 65.46 milliseconds, and QUIC has 99.79 milliseconds. This is in line with the previous discussion about jitter that TCP protocol still has a superior level of transmission rate stability compared to QUIC. The results in this experiment also show a pattern of increasing delay duration as the distance between AP and STA increases.

The packet loss ratio data also shows that QUIC has a larger deviation than TCP. The packet loss ratio is a metric that correlates with delay and jitter. Therefore, the packet loss values obtained are in line with expectations and show the same trend as the jitter and delay data. However, unlike delay, packet loss data does not show a pattern related to the increasing distance between AP and STA. In this experiment, the average value of packet loss for TCP is 0.91%, and QUIC is 1.27%. Meanwhile, the standard deviation value is 1.10% for TCP and 2.08% for QUIC.

After conducting baseline testing on TCP and QUIC with variations in the application type and transmission direction, it can be seen from the analyzed metrics that there are differences in the performance and characteristics of these two protocols. It is also known that the variation of application type and transmission direction does not have an impact in the form of significant metric differences compared to the variation of protocol type (TCP and QUIC). This test also proves that QUIC shows a significant increase in throughput in the distance range of 1-25 meters. However, behind this throughput improvement, QUIC has higher jitter, packet loss, and delay deviation compared to TCP. This is because QUIC is basically a UDP protocol that has been developed to have enhanced TCP features.

C. TCP And QUIC On Shared Connection

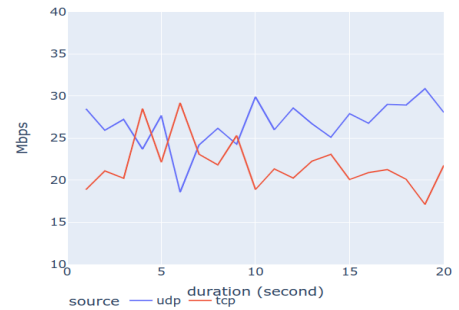


Fig. 5. TCP With UDP

1) *TCP With UDP*: Figure 5 is the average throughput obtained by TCP and UDP. It can be seen that the UDP protocol excels at both distances with an average of 26.70 Mbps. Meanwhile, TCP obtained an average value of 21.87

Mbps. The graph also shows that UDP does not always come on top during bandwidth capture. This is indicated by the TCP throughput value, which exceeds UDP by 4 to 6 seconds at a simulated distance of 5 meters. But afterward, UDP absolutely dominated the bandwidth capture.

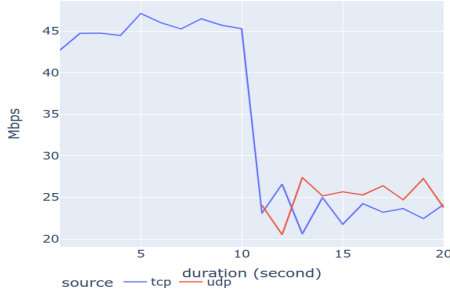


Fig. 6. TCP, then UDP

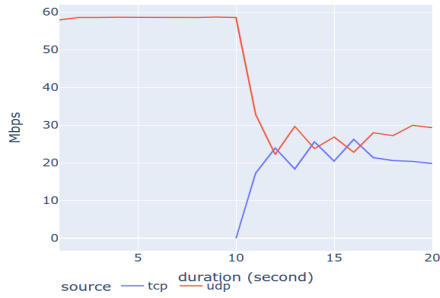


Fig. 7. TCP after UDP

Figures 6 and 7 show the experimental results with varying start times between TCP and UDP. From both figures, it can be seen that there is a struggle for bandwidth between the two protocols. However, in the end, UDP always has a higher average throughput when both protocols start simultaneously. This shows that UDP packets have a tendency to take precedence in the network.

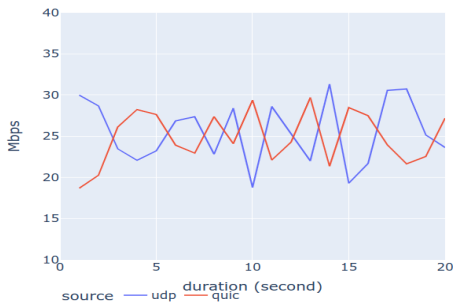


Fig. 8. QUIC With UDP

2) *QUIC With UDP*: Figure 8 shows the simulation results of QUIC and UDP protocols running simultaneously. Unlike the TCP and UDP scenarios, QUIC tends to be able to keep up with the throughput of UDP traffic. This can be seen from the alternating pattern. The average throughput difference between

QUIC and UDP is also thinner when compared to TCP and UDP. In this simulation, UDP gets a throughput value of 25.51 Mbps at a distance of 5 meters. Meanwhile, QUIC gets a value of 24.88 Mbps.

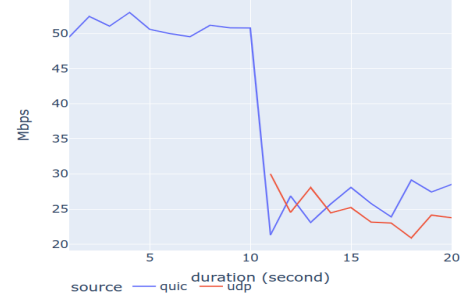


Fig. 9. QUIC, then UDP

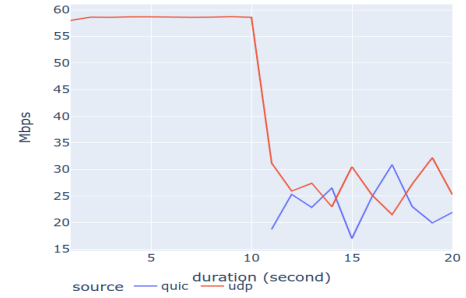


Fig. 10. QUIC after UDP

Figures 9 and 10 show the experimental results with varying start times between QUIC and UDP. From the graphs, it can be seen that the superior protocol type is determined by the protocol that first runs traffic on the network. This can be seen in the scenario where QUIC runs first, UDP runs 10 seconds later then QUIC still dominates the bandwidth on the network. The opposite also happens when the network traffic is first fulfilled by the UDP protocol. This is in line with expectations, given that the QUIC protocol is also detected as UDP in the layer below the fourth layer.

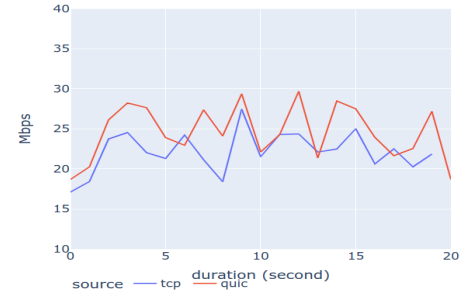


Fig. 11. QUIC With TCP

3) *QUIC With TCP*: Figure 11 shows the simulation results of TCP and QUIC running simultaneously. It can be seen that at both distances, QUIC shows a higher throughput value

than TCP. QUIC has an average throughput of 25.47 Mbps, while TCP gets a throughput value of 22.18 Mbps. The graph also shows that the bandwidth struggle is almost always won by QUIC every time. Furthermore, figures 12 and 13 show the scenario where QUIC runs first and TCP runs first. Both scenarios show that QUIC still has the average throughput value that outperforms TCP.

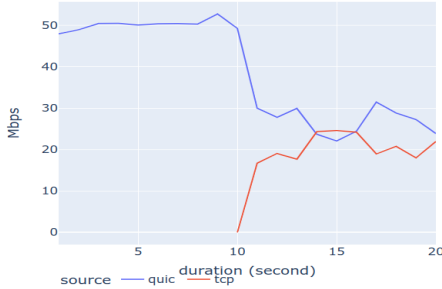


Fig. 12. QUIC, then UDP

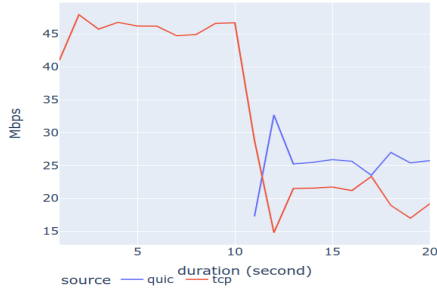


Fig. 13. QUIC after UDP

The conclusion that can be drawn in the experiments in this section is that in the struggle for bandwidth QUIC shows its advantages because it uses UDP technology as its basis. This is shown in the advantage when running concurrently with TCP, which is the same as UDP. In addition, when running concurrently with UDP, QUIC can balance the bandwidth contention so that it has a throughput value that is not much different from UDP.

V. CONCLUSION

Based on the experiments that have been carried out, the following conclusions are obtained:

- 1) From the first experiment, the results show an increase in throughput compared to TCP with an average increase of 7.53% in independent test scenarios.
- 2) The greater the distance between the AP and STA, the smaller the difference in throughput values between the three protocols, with a decrease in the value of the difference between protocols by 1% at each decrease in distance.
- 3) From the first experiment, it was also found that QUIC has a higher deviation of delay, jitter, and packet loss

values compared to TCP, with an increase of 1.5% delay, 1% jitter, and 1.27% packet loss ratio compared to TCP.

- 4) The second experiment was to observe the connection fairness of the two protocols when running simultaneously. In this experiment, it was found that the QUIC protocol would always manage to get an average of 60.8% of the bandwidth when running with TCP, regardless of the starting order. This is exactly the same scenario as UDP running with TCP.

VI. FURTHER STUDY

Based on the results of the research and analysis conducted, some suggestions for further research are as follows:

- 1) Further research can use real network devices and applications that run in a real-world test environment. The number of devices both servers and clients can be increased in order to simulate heavier network loads.
- 2) With the development and increasing use of 5G mobile network links, future research should explore the advantages of QUIC when applied to these links.
- 3) Although QUIC is closely related to HTTP/3, there have been many studies that use QUIC for other applications such as IoT application communication via MQTT protocol.

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